

1. 請框出答案. 2. 不可使用手機、計算器，禁止作弊!

1. Find all solution of the given linear system, using the augmented matrix and row operators.

$$\begin{cases} x_1 + 3x_2 - 2x_3 = 5 \\ 2x_1 + 7x_2 - 3x_3 = 13 \\ x_1 + 5x_2 = 11 \end{cases}$$

Answer: $\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -4 \\ 3 \\ 0 \end{bmatrix} + t \begin{bmatrix} 5 \\ -1 \\ 1 \end{bmatrix}, t \in \mathbb{R}$

Solution :

You may use the method from Example 6 in Section 1.4 of the textbook.

augmented matrix: $\begin{bmatrix} 1 & 3 & -2 & 5 \\ 2 & 7 & -3 & 13 \\ 1 & 5 & 0 & 11 \end{bmatrix}$, reduced row-echelon form: $\begin{bmatrix} 1 & 0 & -5 & -4 \\ 0 & 1 & 1 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix}$

The corresponding linear system is

$$\begin{cases} x_1 - 5x_3 = -4 \\ x_2 + x_3 = 3 \end{cases}$$

Here x_3 is a free variable. Let $x_3 = t$, then $x_1 = -4 + 5t$ and $x_2 = 3 - t$. Hence

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -4 \\ 3 \\ 0 \end{bmatrix} + t \begin{bmatrix} 5 \\ -1 \\ 1 \end{bmatrix}, \quad t \in \mathbb{R}.$$

The linear system has infinitely many solutions.

2. Given row echelon form (REF) and reduced row echelon form (RREF). Determine whether each of the following matrices is in
- (A) both RREF and REF, or
 - (B) REF but not RREF, or
 - (C) RREF but not REF, or
 - (D) neither. Briefly justify your answer.

$$(a) \begin{bmatrix} 1 & 0 & 2 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (b) \begin{bmatrix} 1 & 3 & 0 & 2 \\ 0 & 0 & 1 & 4 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad (c) \begin{bmatrix} 1 & 2 & 0 \\ 0 & 0 & 0 \\ 0 & 1 & 3 \end{bmatrix}$$

$$(d) \begin{bmatrix} 1 & 4 & -2 \\ 0 & 1 & 5 \\ 0 & 0 & 1 \end{bmatrix} \quad (e) \begin{bmatrix} 0 & 1 & 0 & 3 \\ 1 & 0 & 0 & 2 \\ 0 & 0 & 1 & 1 \end{bmatrix} \quad (f) \begin{bmatrix} 2 & 1 & 3 \\ 0 & 3 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

Answer: (a) A (b) A (c) D (d) B (e) D (f) B

Solution :

特別注意一下，不可能存在有 C 型的矩陣。

3. Prove or disprove that if $AB = O$ (the zero matrix), then $A = O$ or $B = O$.

Solution :

1-3 #21(c) 。